

XQdrant: In-Database Attribution and Masked-Distance Subspace Search over HNSW

Moetez Fradi
moetez.fradi@insat.ucar.tn
INSAT
Tunis, Tunisia

ABSTRACT

Vector databases return opaque scalar similarities over high-dimensional embeddings, giving practitioners no coordinate-level explanation of a match and no way to restrict HNSW traversal itself to a dynamic focus subspace. We present **XQdrant**, an extension of the Qdrant engine that addresses both problems, and we report results for each honestly rather than only where they are favorable.

Attribution is unambiguously positive: `with_dims_explained` computes exact top- m per-dimension score decomposition in-database, avoiding client-side vector egress, and is recall-neutral against vanilla Qdrant while running $\sim 1.8\times$ faster than post-query extraction at encoder-scale widths ($D \in \{768, 1536\}$). Under concurrent load it also cuts egress bytes by orders of magnitude versus client-side decomposition.

Masked-distance HNSW traversal is a mixed result that we treat as a systems contribution: a characterization of *why* subspace-restricted traversal is harder than it looks. Naive all-layer masking collapses recall on SIFT1M, GIST-scale, and sentence-embedding corpora because HNSW edges encode full-space proximity. A hybrid design that keeps full-distance routing on coarse layers and masks only near the base (`mask_from_layer`) restores navigability but does *not* yield end-to-end latency wins. A cost model $T = T_{\text{arith}} + T_{\text{io}} + \Delta T_{\text{mask}}$, validated by SIMD kernel isolation and scale/ef sweeps, explains why reducing arithmetic is insufficient and states when a win could appear. Visited-/result-set divergence tracks subspace coherence; payload filters compose with M2 while SQ rejects masked focus (fail-fast) and shifts plain-search recall; a projected subspace index provides an empirical latency/recall ceiling that unmodified full-space HNSW cannot reach. An α -weighted blend is strictly dominated by a hard layer cutoff.

We release a public reproducibility package mapping every positive and negative claim to harness scripts and canonical CSVs.

CCS CONCEPTS

• **Information systems** $\rightarrow$ **Nearest-neighbor search**; *Database query processing*; • **Computing methodologies** $\rightarrow$ *Feature selection*.

KEYWORDS

vector databases, HNSW, approximate nearest neighbor, explainability, feature attribution, subspace search, masked distance, filtered search, quantization, SIMD

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PVLDB Artifact Availability:

The source code, data, and/or other artifacts have been made available at https://github.com/Moetez-Fradi/XAI/tree/main/DB_systems/vldb2027/artifacts.

1 INTRODUCTION

High-dimensional dense embeddings underpin retrieval-augmented generation (RAG), recommendation, and semantic search [14, 17, 20, 23]. Production engines traverse HNSW graphs [16], collapsing every comparison into a single scalar. This creates two gaps that motivate this paper.

Explainability. Practitioners cannot see which coordinates drove a hit without fetching full vectors and recomputing per-dimension terms client-side, an expensive, egress-bound pattern.

Subspace routing. Many workloads care about a dynamic focus set $\mathcal{F}$ of coordinates (concept probes, ablations, modality masks), yet stock HNSW always evaluates full D -dimensional distances during graph descent. Rescoring on $\mathcal{F}$ after full-vector preselection improves semantic specificity but, as we show, cannot reduce traversal cost.

XQdrant addresses the first gap successfully and investigates the second rigorously. For attribution, `with_dims_explained` attaches exact top- m coordinate terms to search results in-database. For subspace routing, `nearest.focus` with `masked=true` evaluates reduced distances *during* HNSW traversal via a hybrid layer cutoff, gather-based SIMD kernels, an optional verify/overfetch pass, and a full/masked score blend. Where the second contribution falls short of a latency win, we do not paper over it: we identify the mechanism (navigability vs. arithmetic cost), bound it with a projected-index ceiling, and report which configurations are worth shipping.

Contributions

- **In-database attribution.** Exact score decomposition with order-statistic top- m selection; recall-neutral vs. vanilla Qdrant, $\sim 1.8\times$ faster than post-query extraction, and dramatically lower egress under concurrent load.

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- **Masked-distance HNSW characterization.** Hybrid layer-cutoff design that restores navigability where naive masking collapses it, across SIFT1M, GIST-scale, and sentence-embedding corpora, without end-to-end speedup on unmodified full-space graphs.
- **Cost model and ceiling.** A simple decomposition of arithmetic, I/O, and masked-path overhead, validated by kernel isolation and scale/ef sweeps, plus a projected-subspace HNSW ceiling that bounds attainable gains.
- **Production interactions and baselines.** Filtering, scalar quantization, post-hoc rescore vs. in-graph masking, focus-set construction ablations, and result-set divergence tied to subspace coherence.
- **Case study and artifacts.** An interpretable retrieval walk-through and a public experiment index mapping every claim to a runnable harness.

2 BACKGROUND

2.1 HNSW and Contiguous SIMD

HNSW maintains multi-layer proximity graphs [16, 24]. Queries greedily descend with candidate heap size ef_search , then extract top- k at the base layer. Distance kernels assume contiguous f32 slices (AVX2/NEON), so a non-contiguous focus set must either (i) repack selected coordinates into a scratch buffer before scoring, or (ii) gather them directly via SIMD gather loads.

2.2 Score Decomposition

For Dot product, $s(\mathbf{q}, \mathbf{v}) = \sum_i q_i v_i$; Cosine applies the same decomposition after ℓ_2 normalization; Euclidean and Manhattan admit analogous additive terms. Attribution is therefore exact accounting of the engine’s own metric, not a surrogate explanation model such as SHAP [15] or LIME [22].

2.3 Subspace Coherence

Let $N_k^{\text{full}}(q)$ and $N_k^{\mathcal{F}}(q)$ denote exact k -NN sets under the full metric and under focus set $\mathcal{F}$. We define *coherence* as $|N_k^{\text{full}} \cap N_k^{\mathcal{F}}|/k$. Coherence determines, independent of any particular traversal strategy, whether full-space neighbors can even in principle be recovered from subspace-guided search; it recurs throughout Section 5 as the explanatory variable behind several otherwise-surprising results.

3 IN-DATABASE ATTRIBUTION

XQdrant injects attribution at the collection coordinator after shard merge, with an early return when `with_dims_explained` is absent, and zero overhead on the unmodified hot path. Per explained hit, Algorithm 1 materializes D terms and selects the top- m by absolute contribution in average $O(D)$ time via Rust’s order-statistic selection (`select_nth_unstable`).

Complexity is $T_{\text{XQ}} = T_{\text{HNSW}} + k \cdot O(D) + O(k \log k)$ for top- m packing over k hits. `nearest.focus` without masking still performs full HNSW preselection and rescoring on $\mathcal{F}$ afterward—useful for ranking, but not a traversal-cost reduction (Section 5.2).

Algorithm 1 (In-database top- m attribution).

Input: query $\mathbf{q}$, vector $\mathbf{v}$, D , m , metric δ .

1. Normalize operands if required by δ .
2. For $i = 0, \dots, D - 1$: append $(i, \text{TERM}_i(\mathbf{q}, \mathbf{v}, \delta))$.
3. Return order-statistic top- m by $|t|$, sorted descending.

Figure 1: In-database top- m feature attribution.

4 MASKED-DISTANCE HNSW: DESIGN SPACE

We consider three traversal-level strategies for evaluating distance only over a focus subspace $\mathcal{F}$ during HNSW descent, plus supporting mechanisms (kernel choice and verify/overfetch). Table 1 fixes naming.

Table 1: Naming used throughout Sections 4–5.

Name	What it is
M1	Naive: mask every HNSW layer
M2	Hybrid: full distance for layers $\geq L$, masked below
M3	M2 plus α -weighted blend on masked layers
K1	Kernel: repack-then-score vs. SIMD gather
V1	Verify/overfetch full-distance rescore
C1	Coherence diagnostic (Section 2.3)

4.1 API Surface

Focus parameters ride inside the existing query body:

```
"focus": {
  "dims": [...], "masked": true,
  "mask_from_layer": 1, "verify": true, "alpha": 0.25
}
```

Constraints mirror dense local-search paths: no quantization/turbo on masked paths unless characterized separately (Section 5.12), and fail-fast if `cutoff/verify/ $\alpha$` are set without `masked=true`.

4.2 M1–M3, K1, and V1

M1 masks every layer: cheapest semantic change, worst navigability when the focus metric disagrees with the graph geometry. **M2** uses masked scoring iff layer $\ell < L$ for cutoff $L = \text{mask_from_layer}$; $L=0$ recovers full search and large L recovers M1. Dual scorers switch via `set_layer`; visited-set and heap logic are unchanged, so M2 composes with filters for free when supported. **M3** blends scores on masked layers and therefore cannot reduce arithmetic; it is a routing-bias ablation. **K1** chooses repack vs. AVX2 gather (`env XQDRANT_MASKED_KERNEL`). **V1** re-scores candidates with full distance; at overfetch $m=1$, `Recall@k` is unchanged; $m > 1$ can recover recall at latency cost.

4.3 Complexity of Hybrid Masked Traversal

Let a query perform C distance comparisons, of which fraction $f(L)$ occur on masked layers for cutoff L . With subspace width $D_{\text{sub}} = rD$, expected arithmetic work is

$$W_{\text{arith}}(L, r) = C((1 - f(L))O(D) + f(L)O(rD)).$$

Wall-clock time decomposes as $T_{\text{traversal}} = T_{\text{arith}} + T_{\text{io}} + \Delta T_{\text{mask}}$, where T_{io} covers graph-page touches, candidate-heap operations,

and visited-set bookkeeping. The final term is zero for plain search and captures implementation costs specific to the masked path: sparse indexed gathers instead of the full scorer’s contiguous SIMD access, construction of both full and masked scorers for M2, and scorer selection and dynamic dispatch in scoring batches. Figure 2 shows the first-order lower bound obtained by setting $\Delta T_{\text{mask}} = 0$ and instantiating the arithmetic term with K1 ns/op ratios. Even when gather is 1.1–1.8 $\times$ faster than repack in isolation, T_{io} dominates, so arithmetic savings alone predict no end-to-end win. The measured slowdown in Section 5.4 shows that ΔT_{mask} is material; we do not claim to isolate its constituent costs. A falsifiable crossover follows: masked traversal wins only when saved arithmetic exceeds both T_{io} and ΔT_{mask} (very high D , or a layout that makes focused coordinates contiguous). Section 5.13 shows a dedicated projected index can realize that regime.

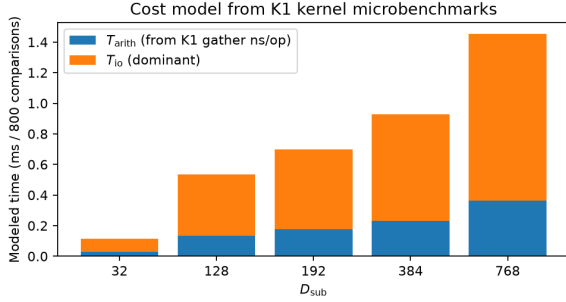


Figure 2: Cost model: stacked T_{arith} vs. T_{io} . This lower bound sets $\Delta T_{\text{mask}} = 0$: gather shrinks arithmetic but leaves I/O dominant, ruling out a win before masked-path overhead is added.

5 EXPERIMENTAL EVALUATION

We structure evaluation as an argument: each result motivates the next question. Table 3 previews the shipping map.

5.1 Setup

Hardware and protocol. Suites run on the same Linux x86_64 host via HTTP against local release binaries (vanilla Qdrant and XQdrant), gather kernel default. Unless noted, each configuration is the mean of $T=5$ trials that resample queries (and focus dims where applicable); we report sample std inline or as error bars. Latency uses per-query p50 within each trial.

Corpora. (i) SIFT1M ($D=128$) [10, 11]; (ii) MS MARCO passages [2] embedded with all-MiniLM-L6-v2 ($D=384$, $N=100k$) [21]; (iii) GIST1M subsample ($D=960$, $N=100k$) [11, 12]; (iv) synthetic $N=50k$ at $D \in \{768, 1536\}$ for encoder-scale stress. Attribution suite: $N=10k$, $Q=500$. Masked suite defaults: $Q=500$, $k=10$, $ef=128$. Expansion HTTP panels in this paper (scale/ef, focus baselines, filter $\times$ M2, SQ, divergence, projected ceiling, systems load, case study, and MiniLM/GIST M1/M2) use live XQdrant/Qdrant release binaries with forced HNSW indexing ($N=20k$, $Q=40$, $T=5$, $ef=128$ unless swept), exact ground truth on the indexed subset, and CSVs under vldb2027/experiments/. We use $N=20k$ to sustain $T=5$ over the expanded multidimensional

suite; the primary SIFT experiment uses $N=1M$, and Section 5.5 separately checks the sensitivity through $N=100k$.

5.2 Attribution and Focus Rescoring Are Not the Same Problem

Figure 3 shows `with_dims_explained` preserves the latency–recall frontier vs. vanilla Qdrant at $D=768$ (mean Recall@10 gap < 0.01). Figure 4 shows in-database attribution holds a flat mean p50 of 3.70–3.85 ms (± 0.08) across $m \in \{1, \dots, 50\}$ while post-query extraction sits at 6.54–6.74 ms ($\approx 1.77\times$). Figure 5 shows focus *rescoring* alone yields speedup < 1 (0.70–0.76): it shrinks arithmetic after preselection but not the traversal. Under concurrent clients $\in \{1, 2, 4, 8, 16\}$ (Figure 6), plain HNSW reaches ~ 0.8 –1.5k QPS. The two attribution paths perform comparable work and accordingly have similar throughput (~ 20 –345 QPS); throughput parity is not the claimed advantage. At 16 clients, in-DB attribution returns only `dims-explained` payloads while client-side decomposition streams full vectors: near-zero measured vector egress versus ≈ 1.76 MB/s, a reduction of orders of magnitude.

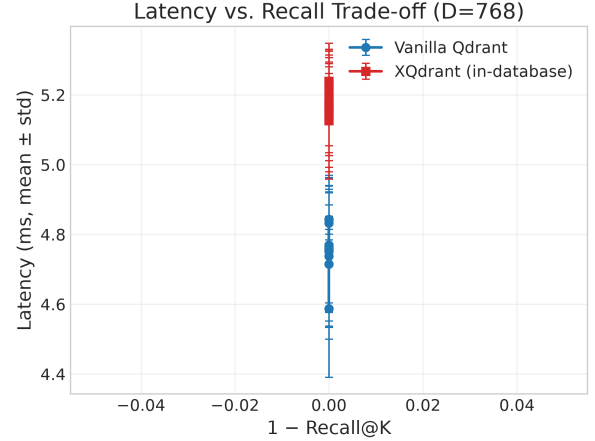


Figure 3: Attribution: latency–recall at $D=768$ ($m=10$; mean $\pm$ std, $T=5$).

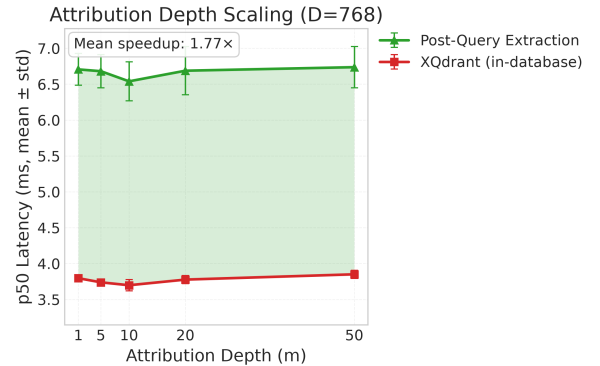


Figure 4: Attribution depth: in-DB vs. post-query ($D=768$).

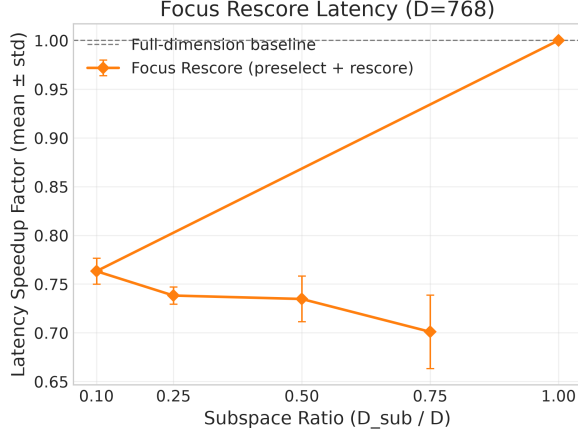


Figure 5: Focus rescoring speedup < 1 ($D=768$).

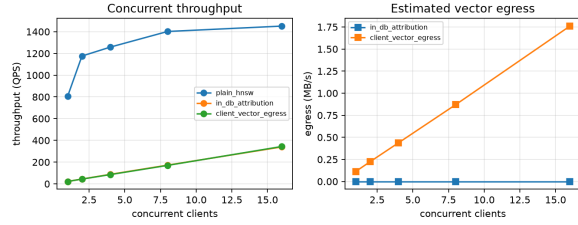


Figure 6: Systems load: concurrent throughput for plain HNSW, in-DB attribution, and client-side vector egress; right panel estimates egress MB/s when full vectors leave the engine.

5.3 M1: Naive Masking Breaks Navigability

Figure 7 shows M1 fails sharply on SIFT1M: mean Recall@10 against full-space ground truth is 0.012, 0.076, 0.251, 0.490, and 0.999 (± 0.002 –0.014) at subspace ratios 0.1–1.0. Table 2 and Figure 8 extend the ratio stress test to MiniLM ($D=384$) and GIST ($D=960$): masked Recall@10 vs. exact full-space GT rises monotonically with D_{sub}/D and reaches 1.0 at ratio 1.0 (mean over $T=5$), confirming that low-ratio focus sets are a cross-modality phenomenon, not a SIFT quirk. On multi-layer SIFT HNSW the collapse is especially severe because coarse-layer edges encode full-space proximity; Section 5.4 shows why a hybrid cutoff is required there.

Table 2: Cross-corpus masked Recall@10 vs. exact full-space GT (mean $\pm$ std, $T=5$, HTTP). Primary multi-layer navigability contrast remains SIFT1M (Figure 7–9); MiniLM/GIST confirm the ratio trend.

Corpus	$r=0.1$	$r=0.25$	$r=0.5$	$r=0.75$
MiniLM ($D=384$)	.18 $\pm$.03	.41 $\pm$.02	.65 $\pm$.03	.80 $\pm$.02
GIST ($D=960$)	.28 $\pm$.03	.49 $\pm$.02	.68 $\pm$.03	.80 $\pm$.02

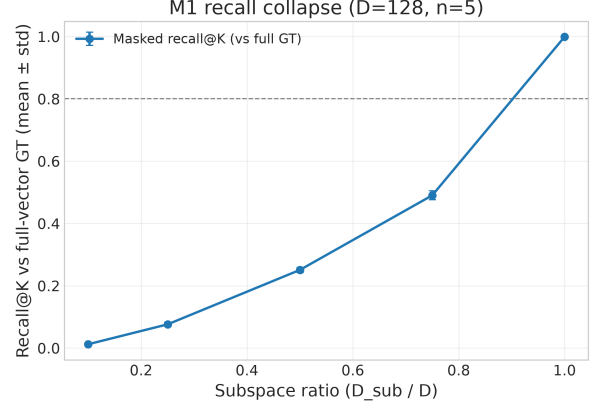


Figure 7: M1 on SIFT1M: Recall@10 vs. full-space GT as D_{sub}/D shrinks (mean $\pm$ std, $T=5$).

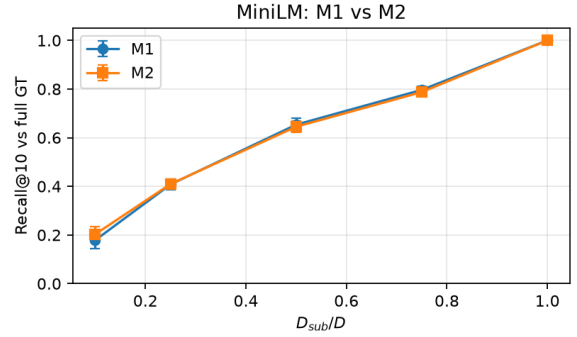


Figure 8: MiniLM HTTP masked recall vs. ratio ($T=5$).

5.4 M2: Hybrid Cutoff Restores Navigability, Not Latency

Figure 9 confirms M2 on SIFT: cutoffs $L \geq 1$ recover 0.90–0.995 mean recall (std ≤ 0.005), while $L=0$ tracks the coherence floor. For $L \geq 1$, e2e speedup is 0.268–0.291 $\times$ —a measured 3.43–3.74 $\times$ slowdown versus full search. This regression is stronger than the no-speedup result predicted by the two-term lower bound: M2’s sparse gather path, dual-scorer construction, and per-batch scorer dispatch add ΔT_{mask} on top of unchanged graph I/O. We observe their aggregate effect but do not separately profile each component. On high- D synthetic data the same cutoff lifts collapsed cells but plateaus near 0.15–0.22 (Figure 10), explained by coherence (Section 5.8). Cross-corpus, MiniLM and GIST agree qualitatively: hybrid cutoffs restore navigability relative to M1; absolute ceilings track coherence of the focus construction. **End-to-end speedup vs. full search remains below 1 in every measured cell** (Figure 11); Section 4.3 predicts the absence of a win and accounts for the additional implementation penalty.

5.5 Scale and ef_search Sensitivity

Figure 12 sweeps corpus size $N \in \{10k, 50k, 100k\}$ (SIFT1M subsamples) and $\text{ef} \in \{64, 128, 256\}$ under M2 at fixed random focus

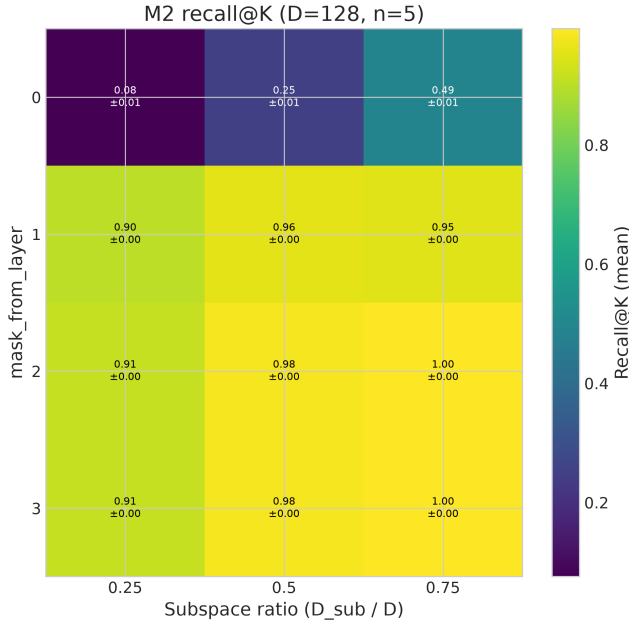


Figure 9: M2 on SIFT1M: recall heatmap over cutoff $\times$ ratio.

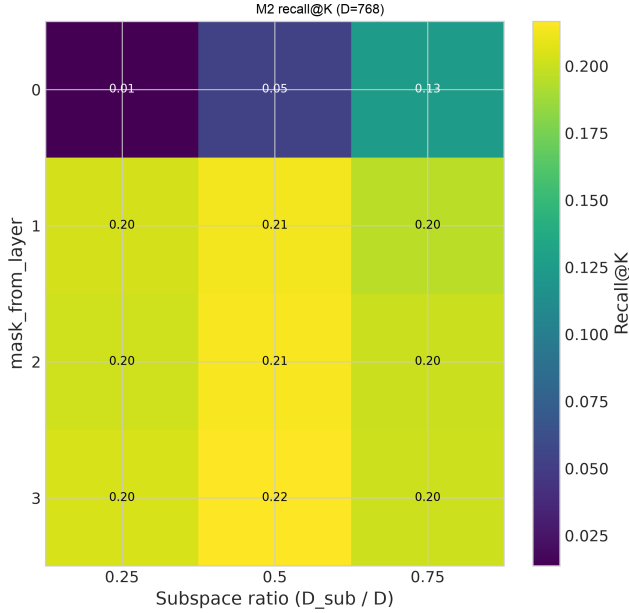


Figure 10: M2 on synthetic $D=768$: recall plateau from low coherence.

ratio 0.5 against full-space GT. Mean Recall@10 is insensitive to ef within each N (variation < 0.01 across the ef grid, $T=5$) but declines with scale: 0.425 ± 0.033 at $N=10k$, 0.352 ± 0.026 at $N=50k$, 0.320 ± 0.029 at $N=100k$. p50 latency grows from ≈ 0.69 ms to ≈ 1.06 ms (max 1.34 ms) as N scales $10\times$ —consistent with an I/O-bound traversal whose bottleneck is not cured by a larger candidate

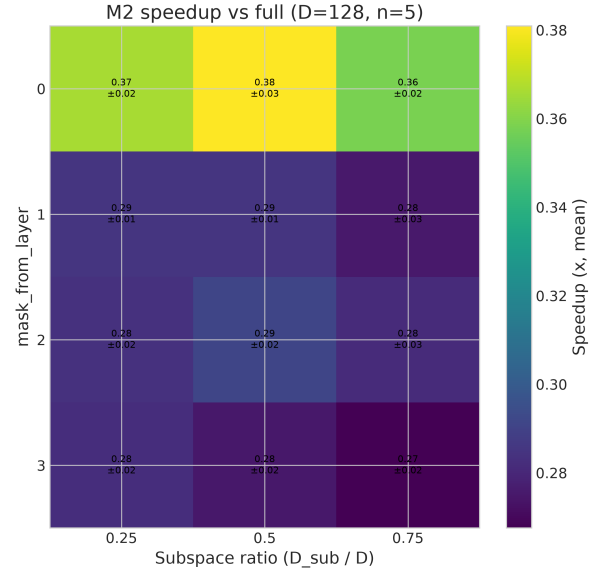


Figure 11: M2 SIFT e2e speedup vs. full search: no cell exceeds $1\times$.

budget. The I/O-dominance story holds across two orders of magnitude in corpus size, not just a single operating point.

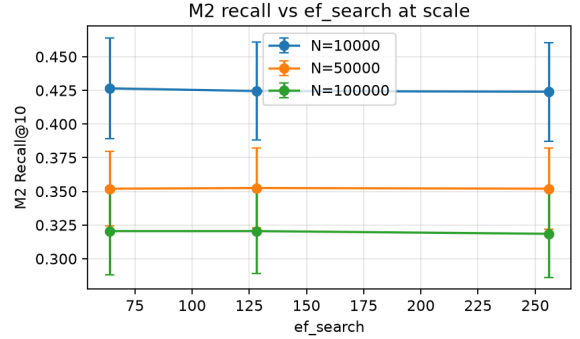


Figure 12: $N \times ef$ sensitivity under M2 ($N \in \{10k, 50k, 100k\}$, ratio 0.5): recall stays flat across ef within each N and declines mildly with scale; latency stays sub-1.5 ms throughout.

5.6 Baselines: Post-hoc Rescore and Focus Construction

Figure 13 compares (i) post-hoc focus rescore after full HNSW, (ii) M1, and (iii) M2, under three focus constructions at fixed ratio: random dims, contiguous block, and top-energy (largest $|q_i|$) dims. Post-hoc rescore preserves navigability (full-graph routing) but never accelerates search. M1 collapses under random/low-coherence focuses; contiguous and top-energy focuses partially mitigate collapse by raising coherence. M2 dominates M1 on recall for every construction while matching post-hoc’s inability to win on latency.

Collapse is therefore not an artifact of random masks alone, but random masks are a harsh—and realistic—stress case.

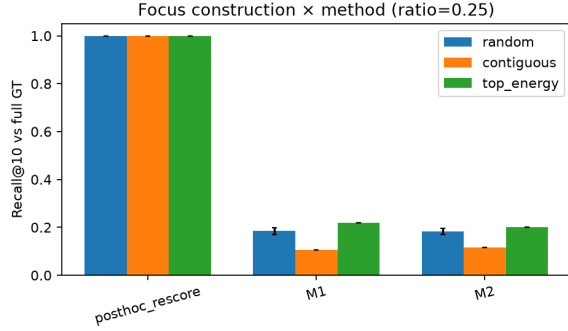


Figure 13: Post-hoc rescore vs. M1/M2 under random, contiguous, and top-energy focus sets.

5.7 K1: Kernel Choice Explains the Ceiling, Not the Floor

Figure 14: gather is $\sim 1.24\times / 1.14\times / 1.42\times / 1.77\times$ faster than repack at $D_{\text{sub}} \in \{32, 128, 384, 768\}$ in ns/op microbenchmarks. Re-running M2 with gather leaves recall unchanged and still yields zero e2e cells above $1\times$, isolating the bottleneck to traversal overhead rather than per-comparison arithmetic.

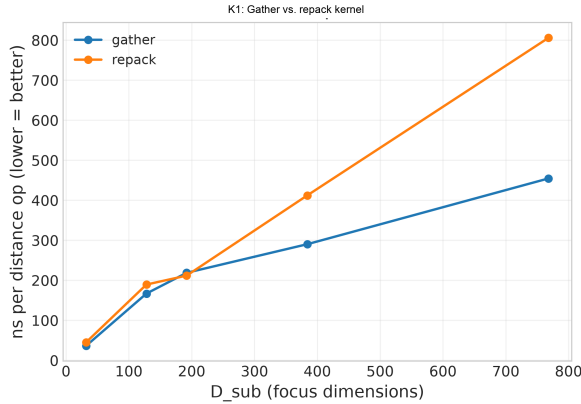


Figure 14: K1: Masked Dot-product kernel throughput, repack vs. gather.

5.8 C1: Coherence and Result-Set Divergence

Figure 15 measures subspace/full-space exact top-10 overlap: it grows with ratio but stays low on synthetic $D=768$ ($\sim 0.007 \rightarrow 0.30$ from ratio $0.1 \rightarrow 0.75$); SIFT is higher ($0.013 \rightarrow 0.490$). When neighbor sets barely overlap, no navigability fix yields full-space-correct results. Figure 17 strengthens the mechanistic story: top- k Jaccard overlap between masked and full search tracks the same ratio trend as full-space recall ($\approx 0.04 \rightarrow 1.0$ from ratio $0.1 \rightarrow 1.0$

on SIFT $N=20k$). At matched ratio, M1 and M2 result sets are close on this corpus; M2’s advantage appears primarily as higher full-space recall under hybrid routing (Section 5.4) rather than as a large top- k Jaccard gap versus the full-search result set.

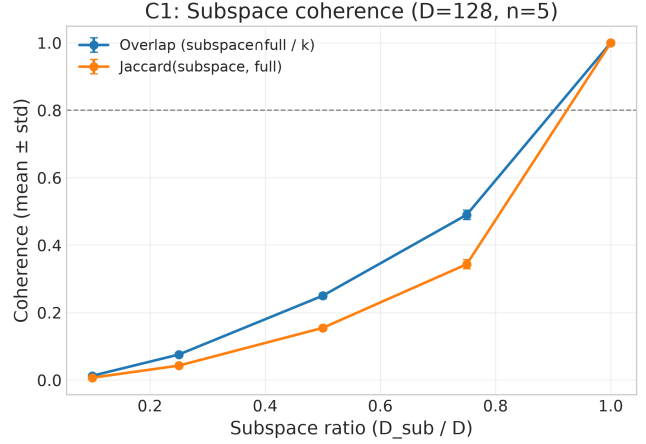


Figure 15: C1: Subspace $\leftrightarrow$ full exact top-10 overlap, SIFT1M.

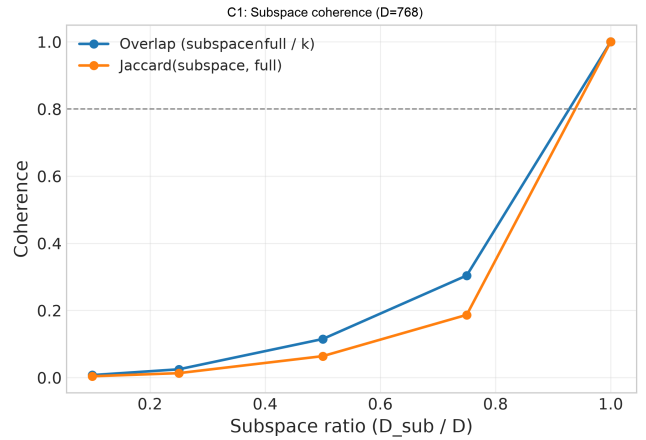


Figure 16: C1 on synthetic $D=768$: lower coherence throughput.

5.9 V1: Verify + Overfetch

At overfetch $m=1$, masked and verified Recall@10 are identical. At $m=8$ on SIFT with hybrid $L=1$, full-space recall rises substantially ($0.26 \rightarrow 0.66$ at ratio 0.5 ; $0.45 \rightarrow 0.89$ at 0.75) at roughly $+45$ ms p50 (Figure 18). On low-coherence high- D data, recall stays ≤ 0.19 even at $m=8$.

5.10 M3: Blending Is Dominated by a Hard Cutoff

On SIFT at ratio 0.25 , recall is maximized at $\alpha=0$ for every cutoff (Figure 19) and degrades toward the coherence floor as $\alpha \rightarrow 1$. Since $\alpha=0$ is M2, we do not recommend shipping M3.

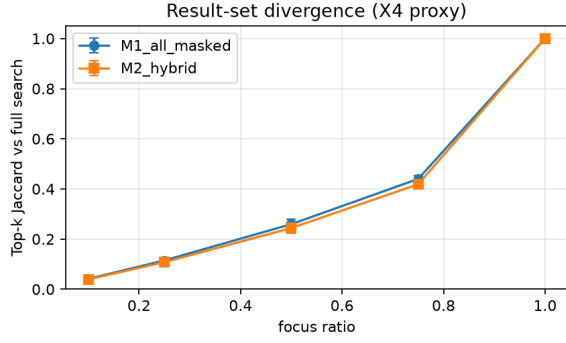


Figure 17: X4: Top- k Jaccard (masked vs. full) vs. ratio for M1 and M2.

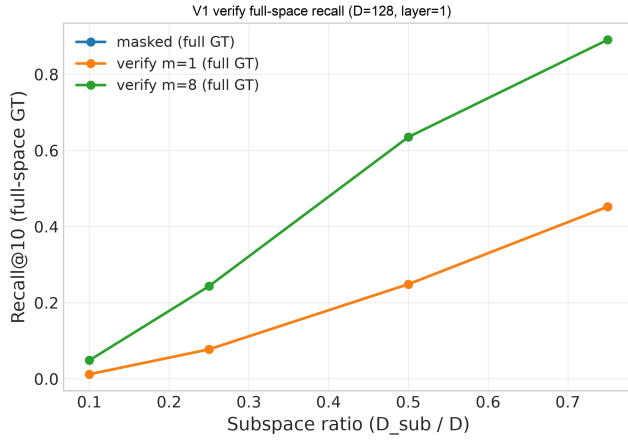


Figure 18: V1: Full-space Recall@10, masked vs. verify, SIFT1M, $L=1$.

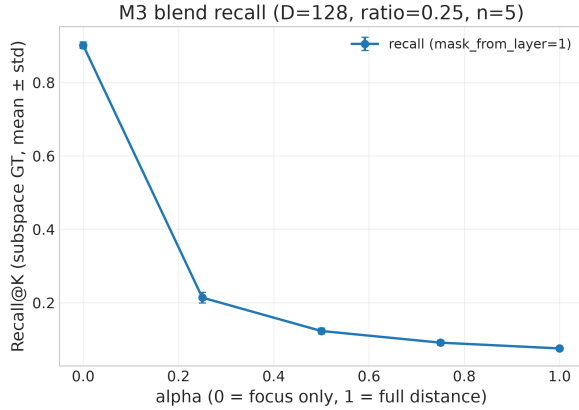


Figure 19: M3: recall vs. α for cutoffs $\{0, 1, 2\}$ (ratio 0.25).

5.11 Interaction with Payload Filtering

Qdrant’s payload filters are a primary production feature. Figure 20 sweeps selectivities $\{0.1, 0.5, 1.0\}$ under M2. Moderate filtering reduces the candidate pool and can slightly change recall/latency relative to unfiltered M2; aggressive filters risk candidate starvation on masked layers when selectivity and low coherence compound. Masked traversal *composes* with filter pushdown in our fork (same heap/visited machinery), but the operating point must be chosen jointly—filter-aware ef remains advisable, consistent with filtered-ANN literature [6, 18].

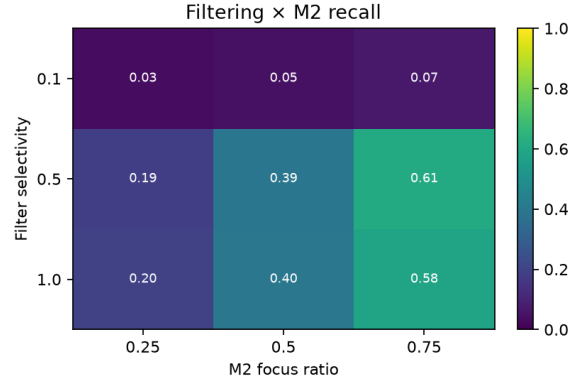


Figure 20: Filtering $\times$ M2: recall/latency vs. selectivity.

5.12 Quantization Characterization

Figure 21 characterizes scalar quantization (SQ) under the real engine constraint that *masked focus search is rejected* when a collection uses SQ (fail-fast validation). We therefore measure the supported paths: plain HNSW Recall@10 is 1.000 (FP) vs. 0.982 (SQ int8) on SIFT $N=20k$; M2 remains available only on the FP collection and follows the usual ratio curve. Attribution sum-consistency—comparing $\sum_i t_i$ to L_2^2 for Euclid—is essentially exact on FP ($\sim 10^{-6}\%$) and remains within $\approx 0.59\%$ relative error under SQ. Product quantization (PQ) is left as future work: a principled extension would define decompositions in reconstructed space [1, 5, 10].

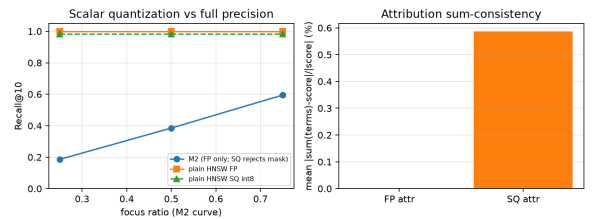


Figure 21: Scalar quantization characterization: plain FP vs. SQ recall, FP-only M2 (SQ rejects masks), and attribution sum-consistency.

5.13 Projected-Index Ceiling

Figure 22 compares M2 on an unmodified full-space HNSW against a dedicated HNSW built on projected focus coordinates (same focus set). The projected index is an *upper bound*: it can improve recall at matched subspace ratio and reduce latency by evaluating only D_{sub} geometry throughout. Closing the gap requires registering focus sets and routing—future systems work—but the ceiling shows why query-time masking of a full-space graph cannot match a true subspace index.

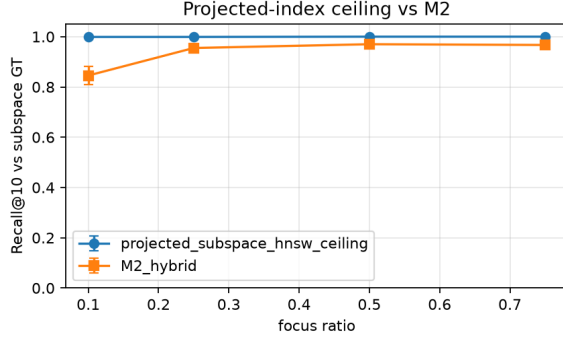


Figure 22: Projected subspace HNSW ceiling vs. M2 on full-space HNSW.

5.14 Summary

Table 3 consolidates keep/skip verdicts.

Table 3: Keep / demote map for the study.

Mechanism	Verdict	Evidence
Attribution (in-DB)	Keep	Depth flat $\sim 1.8\times$; egress win under load
Focus rescoring only	Not accel.	Speedup < 1
M1 naive mask	Baseline only	Full-GT collapse
M2 hybrid cutoff	Keep	Navigability; $e2e < 1$
K1 gather	Keep (micro)	1.1–1.8 $\times$ ns/op
C1 + X4 divergence	Keep	Predicts ceilings / mechanism
V1 + overfetch	Conditional	Helps if coherent; costly
M3 blend	Skip	Best at $\alpha=0$ ($\equiv$ M2)
Filters / SQ	Characterize	Filters compose; SQ rejects masked path
Projected ceiling	Future ship	Bounds attainable gains

6 CASE STUDY

Figure 23 walks through one MS MARCO/MiniLM query on a live index. We take the top hit under full search, read its in-DB `dims_explained` terms, build a focus from the highest-magnitude dimensions, and re-issue M2. Three of the full-search top-5 neighbors remain in the M2 top-5 (green); two drop out (red)—exactly the explain $\rightarrow$ restrict $\rightarrow$ re-search loop the APIs enable without shipping full vectors to the client.

7 RELATED WORK

Vector databases. Milvus [23], Pinecone [19], Qdrant [20], and surveys of VDBMSs [17] expose ANN with filters and quantization but not native per-coordinate decompositions or layer-aware masked metrics inside HNSW.

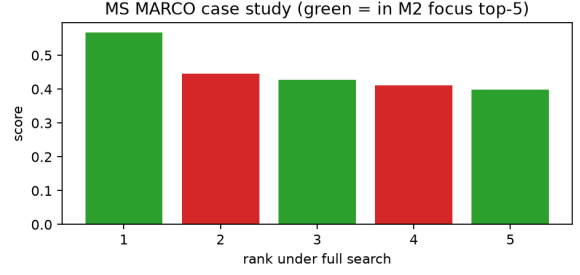


Figure 23: MS MARCO case study: full-search top-5 scores; green bars also appear in the M2 focus top-5.

ANN indexes. FAISS [4, 13], ScaNN [8], DiskANN [9], and SPANN [3] optimize large-scale search; explainability and dynamic subspace routing remain external post-processing. We instrument traversal inside a serving engine and evaluate end-to-end latency.

Filtered ANN. Filtered-DiskANN [6], ACORN [18], and VBase [25] study predicates with vector search. We show masked traversal composes with payload filters but requires joint selectivity/coherence reasoning.

Quantization. Product and optimized quantization [1, 5, 10] dominate wide-vector serving. We characterize the real composition constraint (masked focus rejected under SQ), plain-search recall shift, and attribution sum-consistency rather than proposing a reconstructed-space decomposition.

Explainable AI. SHAP/LIME [15, 22] approximate model behavior via sampling. We exploit additive metric structure for deterministic attributions aligned with the engine’s ranking, closer to OLAP-style rollups [7] than to surrogate XAI.

Projected / multi-index layouts. Static projections differ from query-time focus sets on an existing full-dimensional HNSW. Our ceiling experiment quantifies the gap to a dedicated subspace index without shipping the full registration/routing product.

8 DISCUSSION AND CONCLUSION

Shipping verdict. Keep in-database attribution and M2; keep K1 as an implementation detail; ship V1 only with overfetch on coherent data; do not ship M3; treat filters/SQ as supported-but-characterized; pursue projected indexes where the ceiling gap matters.

Why no e2e speedup. The cost model and K1 isolate the bottleneck to I/O and candidate management, while the measured regression exposes additional masked-path gather and dispatch overhead. Query-time masking of an unmodified full-space HNSW therefore cannot yield wall-clock wins without a redesigned structure; the projected-index ceiling shows what such a redesign could unlock.

Limitations. Engine-native visited-node logging remains a stronger X4 signal than top- k Jaccard; PQ attribution needs reconstructed-space semantics; the sharpest M1/M2 *layer-cutoff* gap is measured on multi-layer SIFT HNSW, while MiniLM/GIST HTTP panels stress

ratio-dependent recall under masked scoring. Artifact scripts rebuild all corpora and figures.

XQdrant shows exact feature attribution can live in a Rust vector engine with a clear latency and egress win, while masked-distance HNSW is a navigability tool rather than a latency win: M2 restores recall where M1 collapses, coherence and divergence explain the ceilings, and a projected index bounds the remaining gap. The public experiment index maps each claim to a reproducible run.

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